

KOTTAPALLI VAMSI

Backend & AI/ML Engineer

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SUMMARY

CSE undergraduate specializing in backend engineering and AI/ML systems, with hands-on experience building production-grade RAG pipelines and full-stack applications using Node.js, FastAPI, PostgreSQL, and LangChain. Delivered end-to-end AI-integrated systems from clinical documentation platforms to real-time WebRTC apps, recognized across national and international hackathons.

EDUCATION

Anil Neerukonda Institute of Technology & Sciences, Visakhapatnam

2023 – 2027

B.Tech in Computer Science and Engineering — CGPA: 8.93/10

Sasi Junior College, Tanuku

2021 – 2023

MPC — 97.7%

EXPERIENCE

Data Analysis Intern | APSSDC – Govt. of Andhra Pradesh

May 2025 – July 2025

- Performed EDA on real-world datasets using NumPy and Pandas; built a weather forecasting analysis project.
- Cleaned and preprocessed multiple real-world datasets, applying data preprocessing and visualization techniques to improve analysis accuracy.

PROJECTS

Clinical Note Intelligence Platform | [GitHub](#)

May 2026 – Present

React, Python, Node.js, FastAPI, Whisper, spaCy, PostgreSQL, Prisma, Redis, Ollama

- Architected a full-stack clinical documentation system with multi-role RBAC (Doctor, Nurse, Receptionist, Admin), JWT refresh rotation, and Redis-backed sessions.
- Engineered a multi-stage audio AI pipeline (Whisper + medical correction layer + speaker diarization), targeting <5% manual correction vs. 20–40% industry baseline.
- Implemented SOAP note generation, medical NER (spaCy/SciSpacy), drug interaction detection (RxNav), and ICD-10/CPT code suggestion via Ollama-powered LLM reasoning with sentence-level audio traceability.
- Developed longitudinal patient intelligence tracking for BP, glucose, and chronic disease trends across visits, auto-generating “What changed since last visit?” summaries.

Medical Literature Research Assistant | [GitHub](#)

Feb 2026 – Mar 2026

Python, FastAPI, LangChain, FAISS, Ollama, Llama, React.js

- Designed hybrid retrieval (BM25 + FAISS) with score fusion (0.6/0.4 weighting), achieving 78.5% precision vs. 65% for BM25 alone.
- Engineered a multi-stage RAG pipeline spanning ingestion, 384-dim embeddings, hybrid retrieval, ranking, LLM synthesis, and hallucination detection, processing 5K PubMed abstracts.
- Implemented safety-critical features: query classification (emergency/diagnosis/treatment), 5-factor confidence scoring, and citation validation for medical accuracy.
- Integrated RxNav and ClinicalTrials.gov APIs for real-time drug interaction and trial retrieval.

Real-Time Messaging & Video Platform | [GitHub](#)

Nov 2025 – Jan 2026

React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS, Socket.IO, Cloudinary

- Developed a full-stack real-time communication platform on the MERN stack with Zustand state management, secured via OTP, JWT, and bcrypt authentication.
- Implemented a low-latency Socket.IO messaging system using an in-memory Map for O(1) socket routing, enabling instant messaging, read receipts, and typing indicators.
- Added peer-to-peer video calling using WebRTC, handling SDP offer/answer negotiation and ICE candidate exchange over WebSockets.
- Designed a media pipeline using Multer for stream parsing and Cloudinary CDN for image and video storage and delivery.

TECHNICAL SKILLS

Languages: C, Java, JavaScript, Python, C++

Frontend: React.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication, Redis

Databases & ORM: MongoDB, PostgreSQL (Prisma ORM), FAISS (Vector DB), BM25, SQL

DevOps & Tools: Docker, Git, GitHub, Postman, VS Code,

AI/ML & Data: LangChain, RAG Pipelines, Data Structures & Algorithms

ACHIEVEMENTS

- Selected for the IEEE National Hackathon Final Round among 3000+ teams nationwide.
- HACK24 Finalist: Delivered a fully functional exam preparation web app within 24 hours.
- National-Level Hackathon (Govt. of India): Developed a web application representing ANITS College.
- Solved 500+ Data Structures & Algorithms problems on LeetCode.